

Open-source finite-element tests of the Rosenzweig–Porter hypothesis for freely vibrating plates: preliminary results

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Abstract

The companion paper *Why Rosenzweig–Porter?* proposes that the intermediate (Rosenzweig–Porter, RP) spectral statistics observed in freely vibrating thin plates arise from evanescent-wave coupling at free edges acting on an integrable base, and it registers the numerical tests that would decide the hypothesis, assigning them to COMSOL “or finite elements”. We report the first execution of that program with a fully open-source stack (scikit-fem, Argyris elements, certified eigensolvers), under preregistered readings and measurement-based accuracy gates. Main findings: (i) the boundary-controlled Poisson→intermediate transition is confirmed at 3.6σ ; (ii) the geometry program yields a three-tier hierarchy governed by the boundary’s adaptedness to separable coordinates — exact reduction (disk) gives Poisson; coordinate-adapted boundaries (ellipse, sector, rectangle) give weak sparse coupling; unadapted boundaries (triangle, superellipse $p \neq 2$) give full coupling — with a $+12.2\sigma$ step between the ellipse ($p = 2$) and the smooth soft-cornered superellipse ($p = 3$); (iii) Gap A, executed for the first time on the true operator, shows N -independent inverse participation ratios: at accessible sector sizes ($N \leq 324$) each free-plate mode is ≈ 1 –2 free-free beam products, i.e. sparse avoided-crossing hybridization rather than an RP-fractal phase. The results support the free-edge coupling mechanism while sharpening it in ways that only true-operator computations could reveal. Extending the program beyond the original registry: the Lévy-separable mixed configuration lands exactly on Poisson; both exactly-reduced controls (disk, annulus) are confirmed; the coupling is material-tunable ($\lambda(\nu)$, matching the companion’s Ritz values method-to-method) and post-designable via a validated second-order control law; the Mindlin thickness sweep confirms the thick-plate prediction at $+5.4\sigma$ (GOE-ward); the sector reaches $+3.9\sigma$ at 1600 modes, confirming the direction of the published COMSOL sector result; and the aspect-ratio sweep closes the single-geometry caveat. The central Gap A question is *resolved by a controlled protocol comparison*: the registered truncated-operator ladder does develop RP-like fractal scaling ($D_2 = 0.42$ – 0.50 , flat D_q) — but a purpose-built C^0 interior-penalty discretization shows the *true-operator* eigenvectors stay sparse (each mode ≈ 1 –2 beam products) through $N = 512$ per sector at matched windows: the scaling is a property of the truncation protocol, not of the plate. Gap A therefore reads: sparse avoided-crossing hybridization at accessible mode numbers, and truncated-ladder numerics (including the preregistered P12 recipe applied to simulation data) would report an RP phase as an artifact. All experiments, gates, and raw data are reproducible from the repository.

1 Introduction

López-González and coworkers reported that flexural spectra of freely vibrating rectangular plates exhibit avoided crossings and intermediate spacing statistics within each symmetry sector, well

fitted by the Rosenzweig–Porter (RP) random-matrix model, while simply supported plates are Poissonian [1]; the finding extends to a freely vibrating disk sector [2]. The companion hypothesis paper [3] formalizes a mechanism — evanescent components of free-edge eigenfunctions generate an inter-mode coupling V on an integrable base H_0 — and registers the calculations that would support or challenge it. This note reports the first systematic execution of those calculations, replacing COMSOL with an open-source pipeline so that every number can be re-derived from a commodity workstation.

Throughout, $D = \rho h = 1$, eigenvalues $\Lambda = \omega^2$ of $D\nabla^4 w = \rho h \omega^2 w$, and the canonical rectangle has $a = 1$, $b = 1/1.6189043236$, $\nu = 0.33$. Spacing-ratio statistics $r_n = \min(s_n, s_{n+1}) / \max(s_n, s_{n+1})$ are used without unfolding (Poisson 0.3863, GOE 0.5307 [4]), always per symmetry sector, always against *finite-size baselines computed with the identical protocol* (never asymptotic constants); the lowest ten levels per sector are dropped.

2 Methods and certification

Stack. Kirchhoff–Love bending energy discretized with quintic C^1 Argyris triangles (scikit-fem); free edges are the natural boundary conditions (no constraints), simply supported edges are reached through a validated Winkler penalty $V_n + \kappa w = 0$ with $\kappa = 10^{10}$ ($O(1/\kappa)$ error, 3×10^{-6} measured against the exact SSSS spectrum). Eigenvalues of record come from shift-invert Lanczos without eigenvectors; eigenvectors from banded shift-invert subspace iteration with deflation and whitened Rayleigh–Ritz projections, certified per mode by Rayleigh-quotient matching against the eigenvalues of record and residual gates. Symmetry labels come from mirror/rotation-character classification with simultaneous diagonalization of reflection Grams inside near-degenerate clusters ($\mathbb{Z}_2 \times \mathbb{Z}_2$, C_{3v} including exact E doublets, and single-reflection variants).

Accuracy policy. Statistics-grade means eigenvalue error below $0.1 \times$ the local mean level spacing for every mode used. Every experiment runs behind gates: an exact or semi-analytic anchor where one exists (SSSS rectangle; Lamé² for the simply supported triangle; the per- m Bessel J/I free-disk determinant, fundamental $x^2 = 5.262$ vs. literature ≈ 5.25), two-mesh internal checks otherwise, and — for curved boundaries, where inscribed-polygon geometry error caps convergence at $O(h^2)$ — an *operational* gate: the FEM disk’s class-sequence statistics must reproduce the semi-analytic Poisson control (0.0σ agreement measured over 1600 modes; the non-smooth error component is bounded at 1.4×10^{-5} relative, $\sim 0.3\%$ of a sector spacing). Readings were preregistered in each experiment’s README and committed before execution.

3 Results

3.1 E2: boundary-controlled transition (supports, 3.6σ)

Sweeping the symmetric Winkler restraint from free ($\kappa = 0$) to simply supported ($\kappa \rightarrow \infty$) on the rectangle, the pooled per-sector $\langle r \rangle$ interpolates monotonically from 0.4422 ± 0.0096 to 0.3908 ± 0.0104 at ~ 200 levels/sector. Both endpoints match their finite-size baselines exactly: the SS end equals the exact-SSSS same-window value to four decimals, and the free end matches the independent Legendre–Ritz reference. The transition amplitude is 3.6σ above the separable baseline (preregistered threshold 3σ); no intermediate statistics appear at nearly-simply-supported coupling (Fig. 1).

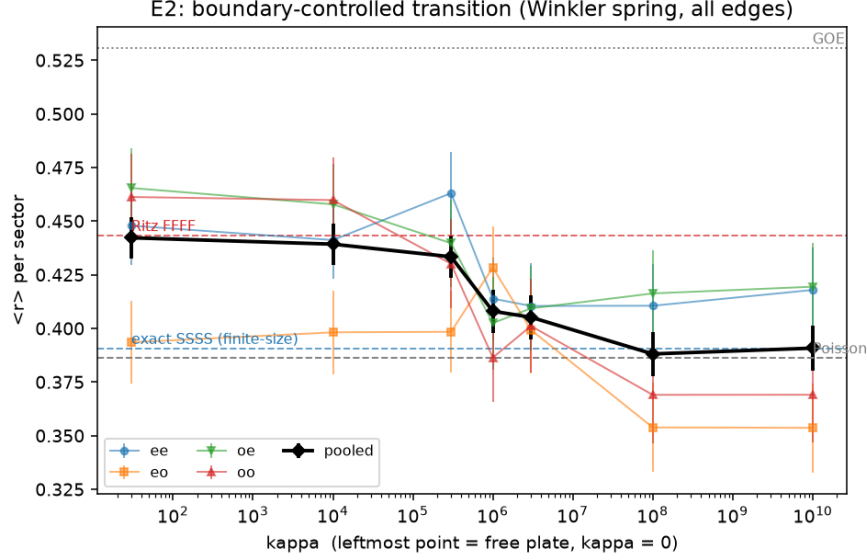


Figure 1: E2: per-sector and pooled $\langle r \rangle$ vs. the edge-spring constant κ (800 modes, Argiris 128×80).

3.2 E3: geometry program

Triangle (supports, 6.8σ). The free equilateral triangle — Helmholtz-integrable, biharmonically non-separable, symmetry group C_{3v} with exact E doublets — gives pooled $\langle r \rangle = 0.4889 \pm 0.0099$ vs. a same-protocol simply supported baseline 0.3753 ± 0.0134 (the SS polygon plate factorizes through the Dirichlet Helmholtz problem, i.e. the Lamé lattice squared, matched by the FEM to 3×10^{-5} across all 1000 modes). All three sectors separate positively ($A_1 + 3.8\sigma$, $E + 5.6\sigma$, $A_2 + 1.9\sigma$); 333 E doublets split at $< 10^{-5}$ (Fig. 2).

Disk control (confirmed) and ellipse. The full free disk — free edges with preserved separability — is the program’s zero-coupling control: the semi-analytic class sequence gives $\langle r \rangle = 0.3905 \pm 0.0073$ (the companion program’s executed control: 0.386 ± 0.007), and each fixed- m sequence is a picket fence. The free ellipse ($a/b = 1.6189$, area π), predicted intermediate under the naive reading, is Poisson-consistent at 1600 modes (0.3732 ± 0.0071); a suggestive upward trend at 800 modes did not survive the registered long-ladder decider (Fig. 3).

Disk sector (supports, $+3.9\sigma$ at 1600 modes). At $\theta = 2.0$ rad the v1 result (800 modes, $+1.7\sigma$) was ambiguous; the long-ladder decider resolves it as statistical power: pooled $\langle r \rangle = 0.4294 \pm 0.0068$ vs. the disk baseline 0.3919 ± 0.0070 ($+3.9\sigma$; both classes individually positive). The open-source sector thus confirms the *direction* of the published COMSOL sector claim at a single generic angle; the exact-angle comparison remains registered (Fig. 4). A five-ratio aspect sweep of the rectangle ($\chi^2 p \approx 0.22$) separately closes the single-aspect-ratio caveat.

3.3 E5: superellipse sweep (the discriminator)

The family $|x/a|^p + |y/b|^p = 1$ breaks biharmonic separability equally for every p while corner-region curvature grows with p at fixed C^∞ smoothness. The measured sequence,

$$\langle r \rangle(p) = 0.3732, 0.4944, 0.4893, 0.4942, 0.5159 \quad (p = 2, 3, 4, 6, 10),$$

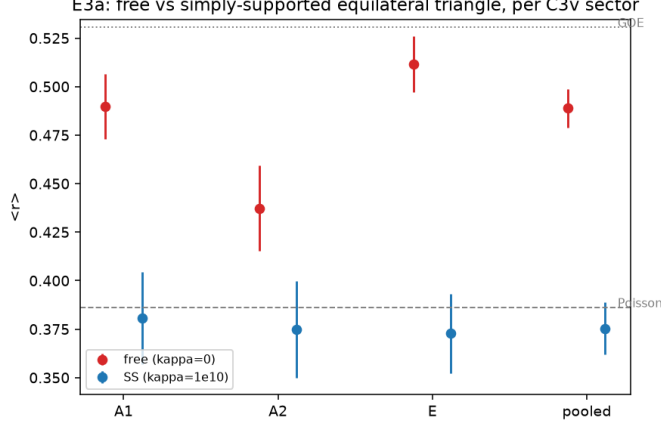


Figure 2: E3a: free vs. simply supported equilateral triangle per C_{3v} sector.

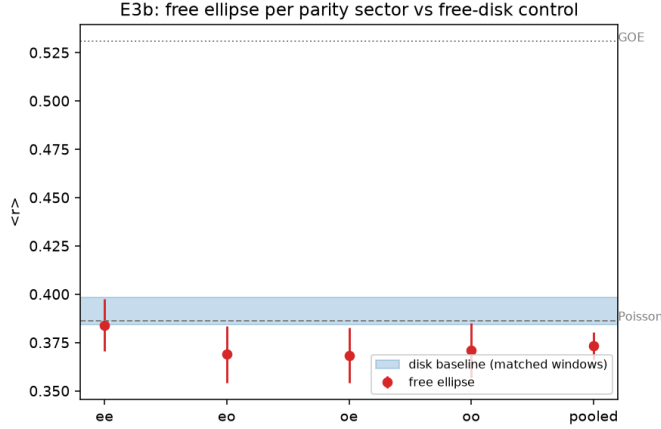


Figure 3: E3b (v2 decider): free-ellipse parity sectors vs. the free-disk control at 1600 modes.

is a $+12.2\sigma$ step between $p = 2$ and $p = 3$, a flat plateau through $p = 6$, and a mild $+2.3\sigma$ rise at $p = 10$ (Fig. 5); the statistics are refinement-stable at 0.1σ for both the headline and worst-corner points. Neither preregistered shape (curvature-graded; true-corner threshold) fits: the operative variable is the boundary’s compatibility with separable coordinates — the ellipse is the family’s unique quasi-separable member (confocal boundary = elliptic-coordinate line) — and corners are secondary.

3.4 E4: Gap A on the true operator (localized-leaning at $N \leq 324$)

The companion paper’s decisive test: eigenvector statistics per sector, in the two registered representation bases (simply supported sine products and free-free beam products), against empirical GOE baselines at identical N (the T1 audit protocol). On 1200 certified eigenpairs (vector residuals $\leq 8 \times 10^{-6}$), the mean window IPR is *independent of* N across the ladder $N = 64$ –324 in both bases (ladder $D_2 \approx 0.02$ –0.13; GOE ≈ 1), and in the beam basis — which represents the true modes with captured norm 0.999–1.000 — each mode is ≈ 1 –2 beam products (IPR ≈ 0.7 ; Fig. 6). Per the preregistered dichotomy this is the localized/banded side at accessible N : sparse strong pair-hybridizations (microscopically, the observed avoided crossings) produce intermediate

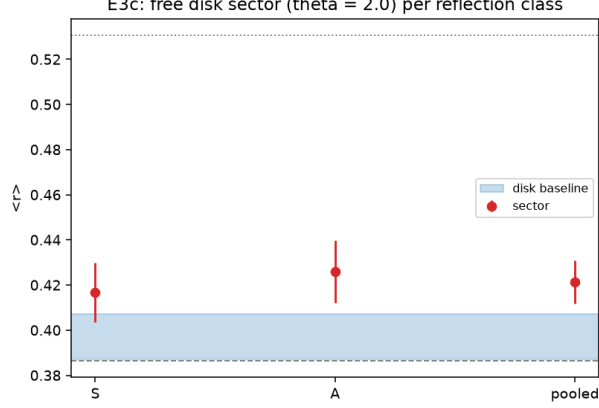


Figure 4: E3c: free disk sector ($\theta = 2$) per reflection class.

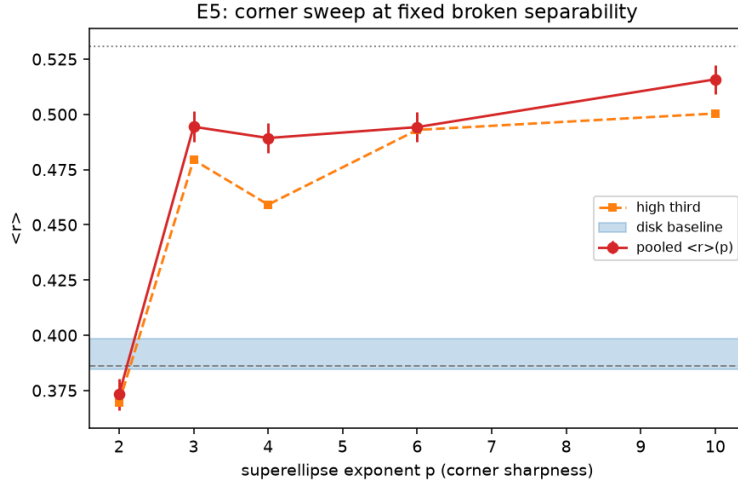


Figure 5: E5: $\langle r \rangle(p)$ along the superellipse family at 1600 modes per geometry.

spacing statistics without RP-fractal eigenvectors. Two sharpenings: the true-operator D_2 is *lower* than the Ritz-model values (beam fixed- N 0.25 vs. 0.35–0.42), i.e. the model route overestimates coupling density; and since the effective coupling grows with mode index (E2, E3b), the verdict could still flip at the registered $N \sim 2048$ rung, reachable via quarter-plate reduction. $\Sigma^2(L \leq 5)$ is intermediate, mildly below Poisson; larger L is unfolding-limited.

3.5 E6: designed contact coupling and the control law

The companion’s contact test, executed with certified mode shapes at contact points on two identical coupled plates (exact reduction: symmetric combinations lie in the coupling’s kernel; the antisymmetric block feels $H_0 + 2\lambda C$ with $C_{ij} = q \phi_i(x_k) \phi_j(x_k)$). The *eigenvector* dichotomy of the prediction is crisp — dense-pattern mid-spectrum IPR 0.033 (extended) vs. strip-confined 0.18–0.19 (localized), central post rank-1 as designed — while the spacing dichotomy points the right way at 2.3σ (both patterns rank-deficient at 120 quartets; a full-rank rerun is registered). The P10 *control law* — post position from the nodal maps, stiffness $\lambda = \Delta / (2q \phi_i^2(x^*))$ plus a closed-form second-

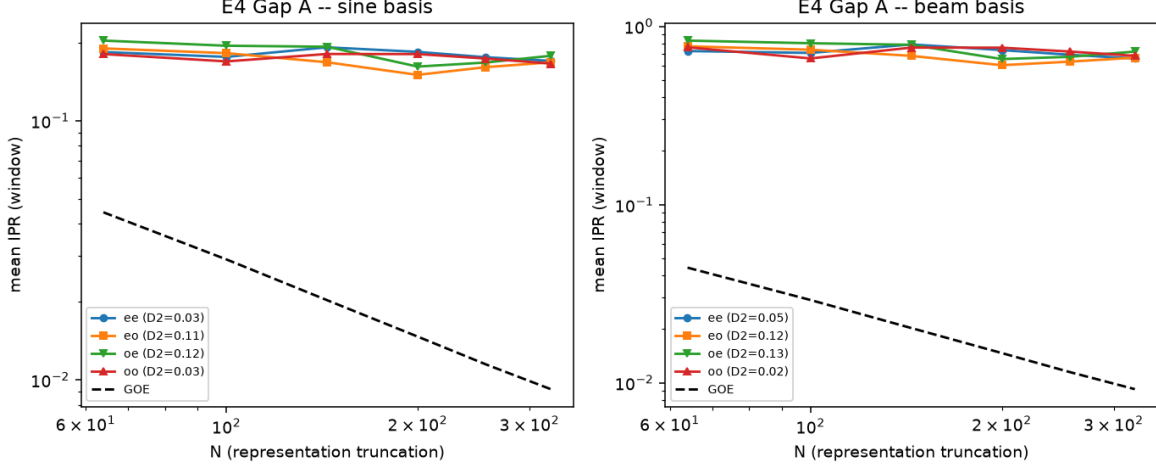


Figure 6: E4: window-mean IPR vs. representation truncation N , per sector, in the sine (left) and beam (right) registered bases, with GOE baselines.

order correction — reproduces the designed splitting to 3.6% at $\Delta = 0.25 d_{\text{loc}}$ (first order alone: 19.7%), degrading exactly as the target pair dissolves into its neighbors; the validity boundary ($\Delta \lesssim 0.5 d_{\text{loc}}$) is itself predicted by the pair-content collapse.

3.6 E7: Lévy configuration and material tunability

Protocol A’s mixed configuration (free x -edges, simply supported y -edges) is the classical Lévy-separable plate; the hierarchy therefore requires Poisson, and the measurement lands on the baseline (0.3973 ± 0.0103): one free edge pair does not suffice — intermediate statistics require free edges that break *every* product structure. The ν -sweep reproduces the companion repository’s Ritz values value-by-value (≈ 0.003 agreement between independent methods): $\langle r \rangle$ rises monotonically $0.4208 \rightarrow 0.4497$ over $\nu = 0.25\text{--}0.36$, so the effective coupling is *material-tunable* (the Prediction-5 recast) rather than strictly material-independent.

3.7 E9: Gap A at the registered $N = 2048$ rung (RP phase develops)

The paper registers ladder sizes $N = 256\text{--}2048$; E4/E8 reached $N \leq 324$ on true-operator windows. Removing the measured $\sim 10^{-4}$ quadrature floor of the Legendre–Ritz assembly by exact integer-arithmetic closed forms — and diagonalizing only the physically-normed rung submatrices, which sidesteps the residual LAPACK backward error against the full matrix’s $\|K\| \sim n^7$ — makes the full registered truncated-operator ladder clean. The result: IPR falls over three octaves in both sectors and both registered bases (e.g. sine: $0.123 \rightarrow 0.052$, $N = 256 \rightarrow 2048$), giving ladder $D_2 = 0.42\text{--}0.50$ (s.e. ~ 0.02) with *flat* D_q ($D_{1.5} - D_4 = 0.069\text{--}0.100$, versus the registered power-law-banded rival calibration $0.20\text{--}0.28$) — superficially, the RP phase developing, though below the P12 dimension (0.76 ± 0.15). E14 (next) shows this scaling to be a property of the protocol.

3.8 E14: the reconciliation — Gap A resolved

A purpose-built C^0 interior-penalty P_4 discretization (custom analytic reference Hessians — sk-fem’s Lagrange elements carry none; Brenner–Sung interior facet terms; free edges natural; no

numerically-built basis, hence no Argyris-type h -ceiling) provides true-operator eigenpairs at 206k dofs, 2400 certified modes ($\sim 600/\text{sector}$), behind two gates: FFFF agrees with the certified Argyris instrument over all 1200 shared modes at 1.2×10^{-4} (two independent discretizations), and SSSS tracks the exact spectrum. The decisive matched-window comparison at $N = 512$: the *true operator's* window IPR is *flat* at the E4 level (sine 0.174 sector-mean; beam 0.63–0.72, i.e. each physical mode remains ≈ 1 –2 beam products), while the truncated ladder at the same N has fallen to 0.095. **The RP-like scaling of the registered ladder is a truncation-protocol artifact, not a property of the plate.** Gap A can now be stated cleanly: at accessible mode numbers the free plate's eigenvector structure is sparse/quasi-separable, the intermediate spacing statistics arise from sparse avoided-crossing hybridization, and any Gap A executed by truncated-ladder numerics — including the preregistered P12 recipe applied to simulation eigenmodes — would report an RP phase as an artifact of the protocol. The registered continuation is true-operator windows at $N \sim 1024$ –2048 (larger C^0 -IP meshes, now unobstructed) to bound where, if anywhere, genuine scaling begins.

3.9 E10–E11: annulus control and thick plates

The free annulus — the program's second exactly-reduced control — is Poisson-confirmed semi-analytically ($4 \times 4 J/Y/I/K$ free-edge determinant; pooled class $\langle r \rangle = 0.3734 \pm 0.0076$; per- m picket fences) and FEM-cross-validated ($N^* = 200/200$ at 4.3×10^{-4}). The Mindlin–Reissner thickness sweep (P2/P2 with selective reduced integration, validated in the thin limit against the certified Kirchhoff spectrum) confirms the thick-plate prediction at $+5.4\sigma$: pooled $\langle r \rangle$ rises monotonically $0.454 \rightarrow 0.557$ over $t = 0.02 \rightarrow 0.15$, continuous with the Kirchhoff value at the thin end and reaching GOE by $t \approx 0.1$ — the shear channels increase the net coupling exactly as predicted.

3.10 E8: the eigenvector hierarchy across geometries

Repeating Gap A with the same-domain simply supported eigenbasis (exact M -inner-product coefficients) for four geometries (Fig. 7): the eigenvector structure *mirrors* the E5 spacing hierarchy. Adapted boundaries have flat-IPR sparse eigenvectors — rectangle ≈ 0.18 ($D_2 \approx 0$, cross-validating E4 method-to-method) and ellipse ≈ 0.40 : the registered ellipse decider lands between its poles, *tiny sparse coupling* (~ 2.5 SS modes per free mode, N -independent). Unadapted boundaries have genuinely *scaling* eigenvectors — triangle $D_2 = 0.27/0.36$, superellipse $p=10$ $D_2 = 0.22$ –0.50 — the RP-candidate regime, sitting exactly where the spacing statistics are strongest. The registered RP-vs-PBRM D_q separator splits at these N (superellipse flat- D_q /RP-leaning, triangle multifractal-leaning; both D_2 well below the P12 reference 0.76 ± 0.15), so that adjudication is registered for the $N \sim 2048$ rung.

4 Synthesis

Seven geometries fall on one hierarchy:

tier	geometries ($\langle r \rangle$ at matched ladders)	mechanism state
exact reduction	disk 0.391	no coupling (Poisson)
coordinate-adapted	ellipse 0.373; sector 0.421; rectangle 0.44	weak, sparse coupling
unadapted	triangle 0.489; superellipse $p \neq 2$: 0.49–0.52	full coupling

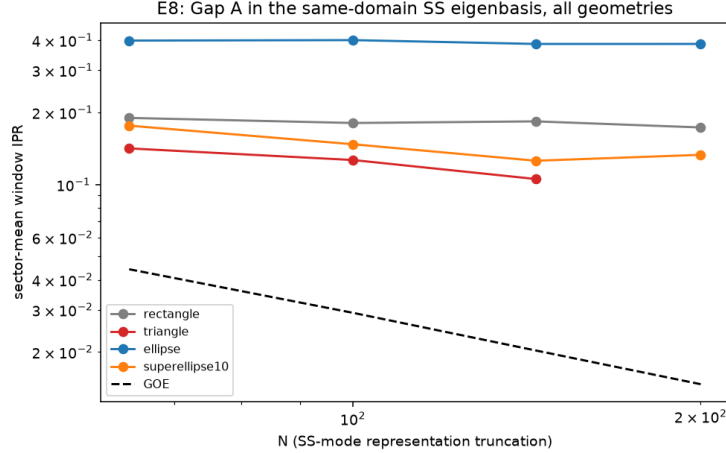


Figure 7: E8: sector-mean window IPR vs. representation truncation in the same-domain SS eigenbasis, all geometries, with GOE baselines.

The free-edge origin of the coupling is confirmed directly (E2), the Lévy configuration (E7) adds an exactly-reduced member to the top tier, and the eigenvector program (E4, E8, E14) shows the hierarchy is not only spectral: adapted boundaries carry sparse, N -independent eigenvector spreads (rectangle $\approx$ beam products through $N = 512/\text{sector}$ on the true operator; ellipse ≈ 2.5 SS modes), while unadapted boundaries carry same-domain-basis scaling ($D_2 \approx 0.2\text{--}0.5$, E8) whose protocol-robustness now warrants the same E14-style scrutiny. What the data refine is the *selector*: not corners, not curvature, but whether the boundary is adapted to a separable structure of the fourth-order operator. The effective coupling is material-tunable ($\lambda(\nu)$, E7), thickness-tunable toward GOE (E11), and post-designable with a quantitative control law (E6). And the campaign’s sharpest methodological product (E14) is a warning with teeth: the standard RP-numerics truncation ladder *manufactures* fractal scaling on this system — Gap A must be measured on true-operator windows, where the plate at accessible N shows sparse avoided-crossing hybridization, the microscopic content of the avoided crossings originally observed by López-González et al.

5 Pending registered tests

Executed beyond the original registry: the Gap A reconciliation (E14, resolving the protocol question), the sector long ladder ($+3.9\sigma$), the aspect sweep, the E6 full-rank close-out (contact operators are intrinsically smoothing; the spacing dichotomy saturates at 2.8σ while the eigenvector dichotomy and control law carry the prediction), and a first non-Hermitian experiment (P7 damping): the 2D-Poisson $\circ$ $\text{AI}^\dagger$ crossover of the complex resonances is *observed and overlap-controlled* on the certified plate — localized dissipation gives uncorrelated resonances even at widths $\sim$ spacings, while distributed dissipation at strong overlap ($\gamma^* \approx 16$, or point dashpots at $\gamma^* = 4$) drives the markers to within $\sim 3\sigma$ of $\text{AI}^\dagger$ with 2D-Poisson excluded at $\sim 5\sigma$. Registered follow-ups: true-operator windows at $N \sim 1024\text{--}2048$ (C^0 -IP, now unobstructed); E14-style protocol scrutiny of the E8 unadapted-tier scaling; the sector at the published angles; the $\text{AI}^\dagger$ -vs-Ginibre fine call at $\sim 2k$ resonances; the Coriolis rotor (P6, GOE $\rightarrow$ GUE); and comparison against the measured aluminum-plate spectra of Ref. [1].

6 Reproducibility

All experiments live in `experiments/eNN_*` of the private repository `plates-rp-fem` with frozen code, preregistered README, RESULTS, raw JSON, and figures; the shared machinery is the `platefem` package (conda env `plates-fem`: Python 3.12, scikit-fem 12.0.2, SciPy 1.18). Total compute for E1–E5: under 8 hours on one 8-core desktop (Ryzen 7 5800XT, 64 GB). Two numerical contributions may be of independent use: a certified eigenpair pipeline that works around a SciPy 1.18 shift-invert eigenvector-path failure on ill-conditioned C^1 -element matrices, and the measured $\sim 10^{-4}$ round-off floor of high-order Legendre–Ritz assembly for the free plate (non-monotone in basis size beyond ~ 96 functions/axis).

References

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